Robust Spec-Level Semantic Matching of Industrial RFP Technical Requirements to OEM Product SKUs

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by

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Approval Certificate

This is to certify that the Project entitled "Robust Spec-Level Semantic Matching of Industrial RFP Technical Requirements to OEM Product SKUs" by Dwayne Fernandes, Soham Ghorpade and Dwayne Fernandes is approved for the partial fulfillment of Major Project - I for the Third Year towards obtaining the Degree of Bachelor of Technology in Computer Engineering.

Project Guide **External Examiner**

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Declaration

I wish to state that the work embodied in this work titled "Robust Spec-Level Semantic Matching of Industrial RFP Technical Requirements to OEM Product SKUs" forms my own contribution to the work carried out under the guidance of Dr.Nataasha Raul at the Sardar Patel Institute of Technology. I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission.

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1 Abstract

Industrial procurement processes require precise matching between Request for Proposal (RFP) technical requirements and Original Equipment Manufacturer (OEM) product specifications. Traditionally, this matching is performed manually, making it time-consuming, error-prone, and highly dependent on domain expertise. While recent advancements in semantic similarity and dense retrieval models have demonstrated strong performance in document-level matching tasks, existing approaches primarily focus on document-to-document similarity or product-to-product matching. They do not address clause-level, specification-wise compliance validation required in engineering procurement workflows. Furthermore, most embedding-based methods ignore numeric constraints, unit consistency, and attribute-level validation, limiting their applicability in real-world technical compliance scenarios.

This research proposes a hybrid, spec-level semantic matching framework for robust RFP-to-product compliance evaluation. The system integrates transformer-based sentence embeddings (Sentence-BERT) for semantic clause matching with a structured numeric and attribute validation engine that performs unit-aware comparison, range checking, and exact constraint validation. The proposed architecture operates in a layered manner: RFP and product catalog specifications are first structured and aligned into a custom dataset; semantic embeddings retrieve top candidate SKUs; a numeric constraint module validates engineering parameters; and a compliance scoring engine produces attribute-wise match/non-match results along with an overall compliance percentage and confidence indicator.

Since no benchmark dataset exists for clause-level RFP-to-SKU compliance matching, a structured dataset was constructed containing aligned semantic, numeric, and categorical engineering attributes such as voltage rating, core configuration, cross-sectional area, insulation type, temperature rating, fire resistance, armouring, and standards compliance. The framework supports robustness testing under incomplete or perturbed specifications, including missing numeric constants, dropped parameters, and unit inconsistencies.

The system is evaluated using retrieval metrics such as Top-K Accuracy and Recall@K, along with spec-level compliance accuracy and robustness degradation analysis. By combining semantic understanding with structured constraint validation, this work bridges the gap between neural semantic similarity models and engineering-grade compliance evaluation. The proposed approach contributes a hybrid architecture, a structured dataset, and an interpretable compliance scoring mechanism tailored for industrial procurement applications.

2 Introduction

Industrial procurement processes rely heavily on Requests for Proposals (RFPs) that specify detailed technical and regulatory requirements for engineering components and systems. These documents commonly contain clause-level specifications such as voltage ratings, cross-sectional areas, conductor configurations, insulation types, temperature limits, fire resistance classifications, armouring requirements, and mandatory compliance standards [1–3]. Accurate interpretation and mapping of these requirements to suitable Original Equipment Manufacturer (OEM) Stock Keeping Units (SKUs) is essential for ensuring operational safety, regulatory compliance, and engineering feasibility.

In current industrial practice, RFP-to-product matching is largely performed manually by domain experts. This approach, while precise, is time-consuming, difficult to scale, and susceptible to inconsistencies when handling large tender volumes. The increasing digitization of procurement workflows has motivated research into automated semantic similarity and retrieval systems capable of assisting tender analysis.

Advances in dense vector representations and transformer-based sentence embedding models, such as Doc2Vec and Sentence-BERT (SBERT), have significantly improved semantic similarity and retrieval performance [6, 12, 15]. Dense retrieval architectures, including late-interaction models such as ColBERT, further enhance efficiency and ranking quality in large-scale search scenarios [9]. These techniques have been successfully applied to product matching and entity resolution tasks in e-commerce and structured data environments [4, 5, 11, 13].

However, existing approaches predominantly operate at the document or entity level and focus on overall similarity scoring or binary matching decisions. Industrial procurement workflows require clause-level, specification-wise compliance validation rather than coarse document similarity. Engineering requirements frequently include explicit numeric constraints, tolerance ranges, and unit-sensitive comparisons. Recent research on quantity-aware retrieval highlights the difficulty of handling numeric intent within dense retrieval models [8]. Purely embedding-based systems lack explicit constraint reasoning mechanisms and may retrieve semantically similar yet technically non-compliant products.

Furthermore, real-world RFP documents often contain incomplete specifications, inconsistent formatting, ambiguous phrasing, or noise introduced during document digitization. Empirical studies demonstrate that embedding models can exhibit performance degradation under noisy or perturbed textual inputs [18, 19], underscoring the importance of robustness-aware evaluation in applied retrieval systems.

Despite significant advances in semantic retrieval and entity matching, there remains a clear research gap in developing a hybrid framework that integrates clause-level semantic matching with structured numeric and attribute-level validation, while also providing interpretable compliance outputs suitable for engineering procurement contexts.

To address this gap, this research proposes a spec-level semantic and constraint-aware matching framework for RFP-to-SKU compliance evaluation. The approach combines transformer-based clause embeddings for candidate retrieval with deterministic numeric validation and structured compliance scoring, enabling interpretable and robust engineering-grade specification matching.

3 Scope of the Project

This project focuses on the research and development of a clause-level semantic and constraint-aware compliance framework for industrial RFP-to-SKU matching. The scope is restricted to technical specification analysis within structured engineering procurement workflows and does not extend to full commercial tender evaluation or enterprise ERP integration.

The system is designed to operate on structured representations of RFP clauses and OEM product catalog data. In the absence of publicly available benchmarks for specification-level compliance matching, a custom dataset is constructed aligning semantic descriptions, numeric engineering parameters, and categorical attributes relevant to industrial cable specifications.

The project encompasses the following functional components:

- **Clause-Level Extraction:** Extraction and segmentation of individual technical clauses from RFP documents to enable granular and attribute-wise analysis.
- **OEM Product Specification Parsing:** Structured parsing and normalization of OEM product catalog specifications to standardize attribute representations for comparison.
- **Semantic Retrieval Integration:** Implementation of transformer-based sentence embeddings (SBERT/MiniLM) for clause-level vector representation and Top-K SKU candidate retrieval.
- **Numeric and Unit Constraint Validation:** Development of a constraint validation engine capable of unit normalization, range checking, inequality comparison ($\geq$, $\leq$), exact matching, and tolerance handling for engineering parameters.
- **Explainable Specification Matching:** Generation of attribute-wise compliance labels (match, partial match, conflict) with interpretable reasoning for each technical parameter.
- **Interpretable Compliance Scoring:** Aggregation of specification-level validation results into a weighted overall compliance percentage along with confidence indicators.
- **Robustness Testing:** Controlled evaluation of system performance under perturbed inputs, including missing numeric values, dropped attributes, paraphrased clauses, and unit inconsistencies.

The implementation is limited to a research-oriented prototype framework developed using structured datasets and simulated perturbations. The scope excludes deployment within live procurement ecosystems, financial or legal evaluation modules, and automated contract finalization. The focus remains on demonstrating the feasibility, interpretability, and robustness of hybrid semantic and constraint-aware compliance matching within an engineering procurement context.

4 Literature Survey

Ref.	Paper Title	Year	Domain	Method / Model	Contribution	Limitations
[1]	Semantic Similarity for Public Procurement Tender Drafting	2024	Public procurement	Doc2Vec, USE, SBERT, RoBERTa	Document-level embedding comparison	No Top-K metrics, no robustness evaluation
[2]	Tender-Sims: Similarity Retrieval System for Public Tenders	2018	Public procurement	Ontology similarity, TF-IDF	Ontology-driven semantic integration	Threshold dependent, ontology constrained
[3]	BidCorpus: A Multifaceted Learning Dataset for Public Procurement	2025	Public procurement	BERT baselines	Procurement dataset construction	No clause-level RFP-SKU analysis
[4]	Entity-Matching System Based on Multimodal Data	2022	E-commerce	BERT, CNN, Random Forest	Multimodal classification	Binary decision only
[6]	Distributed Representations of Sentences and Documents	2014	Text representation	Doc2Vec	Dense document vectors	No structured spec reasoning
[7]	Structure-Aware LM Pretraining for Dense Retrieval	2023	Information Retrieval	Transformer encoder	Structure-aware retrieval improvement	No numeric/unit handling
[10]	Product Matching using Sentence-BERT	2024	E-commerce	SBERT (MiniLM)	Threshold-based semantic matching	No compliance validation
[11]	Sentence-BERT: Sentence embeddings using Siamese BERT net- works	2019	STS	Siamese/Triplet BERT	Improved semantic similarity	Not constraint-aware
[13]	Semantic Product Search for Structured Catalogs	2020	Product IR	DistilBERT, Siamese Nets	Field-wise semantic matching	Retrieval focused only

Ref.	Paper Title	Year	Domain	Method / Model	Contribution	Limitations
[14]	Enhanced Embeddings for Public Procurement	2025	Procure analytics	Domain-adapted SBERT	Domain embedding refinement	No retrieval evaluation
[15]	Sentence Embeddings for Domain IR	2023	Information Retrieval	Domain-adapted SBERT	Comparative embedding study	No robustness study
[16]	Deep Entity Matching with PLMs (DITTO)	2021	Entity matching	BERT/RoBERTa	Transformer pair classification	Binary output only
[17]	LLMs for Public Procurement Recommendations	2024	Procure systems	LLM embeddings	Recommendation framework	No semantic degradation study
[18]	Robust Comparison of Embedding Methods under Noise	2025	Legal retrieval	Domain BERT	Noise-aware evaluation	Sparse annotations
[19]	Robustness of Embedding Models on Noisy Inputs	2025	Information Retrieval	BGE, Arctic-M, Stella	Noise robustness benchmarking	No engineering constraint testing

Table 1: Literature survey of 15 papers

5 Research Gaps

Despite significant advancements in semantic similarity, entity matching, and dense retrieval models, several critical limitations remain in the context of industrial procurement and engineering compliance evaluation.

5.1 Lack of Clause-Level Spec Matching

Existing retrieval and matching approaches primarily operate at the document level or product-to-product level. They generate similarity scores between entire texts but do not perform fine-grained, specification-wise compliance validation. Industrial procurement requires attribute-level comparison such as voltage rating, conductor material, cross-sectional area, insulation type, and standards compliance. Current systems do not provide structured spec-level match/non-match outputs.

5.2 Absence of Public RFP–SKU Benchmark Datasets

There is no publicly available benchmark dataset aligned specifically for clause-level RFP-to-SKU compliance evaluation. Most prior work uses product matching datasets, general semantic similarity datasets, or document retrieval benchmarks. The absence of structured datasets limits reproducibility and standardized evaluation in procurement automation research.

5.3 Lack of Numeric and Unit-Aware Reasoning

Embedding-based retrieval models treat numeric values as plain tokens and fail to perform engineering-aware constraint validation. They do not handle:

- Unit normalization (e.g., kW $\leftrightarrow$ W)
- Range constraints ($\geq$, $\leq$, between)
- Tolerance-based compliance
- Exact numeric validation

This significantly limits their applicability in real-world engineering compliance workflows.

5.4 Limited Robustness to Noisy or Incomplete Specifications

Real-world RFP documents frequently contain incomplete clauses, paraphrased requirements, ambiguous wording, or missing numeric constants. Existing semantic retrieval systems are rarely evaluated under structured perturbations, and robustness degradation is not systematically analyzed.

5.5 Lack of Explainable Compliance Scoring

Most similarity-based systems output raw similarity scores without interpretable explanations. Procurement workflows require:

- Attribute-wise match breakdown
- Conflict identification

- Confidence indicators
- Compliance percentage interpretation

Explainability remains underexplored in structured compliance evaluation.

6 Objectives

Based on the identified research gaps, this work defines the following core objectives:

6.1 Objective 1: Dataset Construction and Spec-Level Semantic Matching Framework

To construct a structured RFP–SKU benchmark dataset aligned at the specification level and to design a clause-wise semantic matching framework capable of generating attribute-level compliance outputs.

This includes:

- Designing structured schemas for RFP clauses and SKU specifications
- Generating aligned semantic, numeric, and categorical attributes
- Implementing transformer-based embeddings for clause retrieval
- Developing a spec-level compliance scoring mechanism

6.2 Objective 2: Numeric-Aware and Robust Matching

To integrate numeric reasoning and constraint-aware validation into the semantic matching pipeline.

This includes:

- Unit normalization and conversion logic
- Range and tolerance-based comparison
- Hybrid semantic + numeric compliance scoring
- Robustness evaluation under incomplete and perturbed inputs

6.3 Objective 3: Explainable Compliance Scoring

To generate interpretable outputs including:

- Attribute-wise match and mismatch identification
- Structured compliance percentage
- Confidence-aware reporting
- Gap analysis for partially matched specifications

6.4 Objective 4: Testing and Comparative Evaluation

To conduct comprehensive performance evaluation using:

- Top-K Accuracy
- Recall@K
- Spec-level compliance accuracy

- Numeric correctness rate
- Robustness degradation analysis

7 Study of Dataset

7.1 Dataset Overview

The proposed spec-level RFP-to-SKU matching framework requires structured datasets that allow direct comparison between technical procurement requirements and product specifications. Since no publicly available benchmark dataset exists for clause-level compliance matching between Request for Proposal (RFP) specifications and product catalogs, a custom dataset was constructed to support semantic matching and attribute-level validation.

The dataset consists of two aligned tables: a Product Catalog dataset containing OEM product specifications and an RFP Specifications dataset representing technical requirements extracted from procurement documents. Both datasets were designed with compatible attribute schemas to enable deterministic comparison and compliance evaluation.

7.2 Product Catalog Dataset

The product catalog dataset represents individual Stock Keeping Units (SKUs) along with detailed engineering specifications. Each row corresponds to a single product entry and contains both descriptive and structured attributes.

The dataset includes the following fields:

- **Identification Attributes:**
  - `sku_id`, `product_name`, `manufacturer`, `category`
- **Technical Specifications:**
  - `voltage_rating`
  - `conductor_material`
  - `core_count`
  - `size_sqmm`
  - `insulation_type`
  - `sheath_type`
  - `temperature_rating`
  - `fire_resistance`
  - `armouring`
  - `standards_complied`
- **Commercial Attributes:**
  - `price_per_unit`
  - `lead_time_days`

These attributes describe the physical, regulatory, and operational characteristics of each cable product and serve as candidate specifications during the matching process.

7.3 RFP Specifications Dataset

The RFP dataset contains structured technical requirements derived from procurement clauses. Each row represents a requirement item that must be matched against suitable products from the catalog.

The dataset includes:

- **Identification Fields:**
  - `rfp_id`, `item_id`, `item_name`, `item_category`
- **Required Technical Attributes:**
  - `voltage_rating`
  - `conductor_material`
  - `core_count`
  - `size_sqmm`
  - `insulation_type`
  - `sheath_type`
  - `temperature_rating`
  - `fire_resistance`
  - `armouring`
  - `standards_required`
- **Constraint Information:**
  - `mandatory_specs`
  - `tolerance_allowed`
- **Semantic Feature:**
  - `spec_text_raw` – Natural language description of the technical requirement used for semantic embedding.

7.4 Feature Alignment Between Datasets

The two datasets were intentionally designed with aligned attribute names and compatible data types. This alignment enables:

- Direct attribute-wise comparison between requirements and products.
- Semantic matching using descriptive specification text.
- Deterministic compliance validation using structured engineering fields.
- Integration of semantic similarity with rule-based evaluation.

Numeric attributes such as voltage rating, core count, and conductor size support quantitative validation, while categorical attributes ensure material compatibility and standards compliance.

7.5 Dataset Design Rationale

The dataset integrates three complementary feature types:

- **Semantic Features** capture clause-level intent using natural language descriptions.
- **Numeric Features** enforce engineering constraints through measurable specifications.
- **Categorical Features** validate material properties and regulatory standards.

This hybrid design enables the proposed system to transition from document-level similarity toward spec-wise compliance evaluation, forming the foundation for the semantic matching framework implemented in Objective 1.

8 System diagram

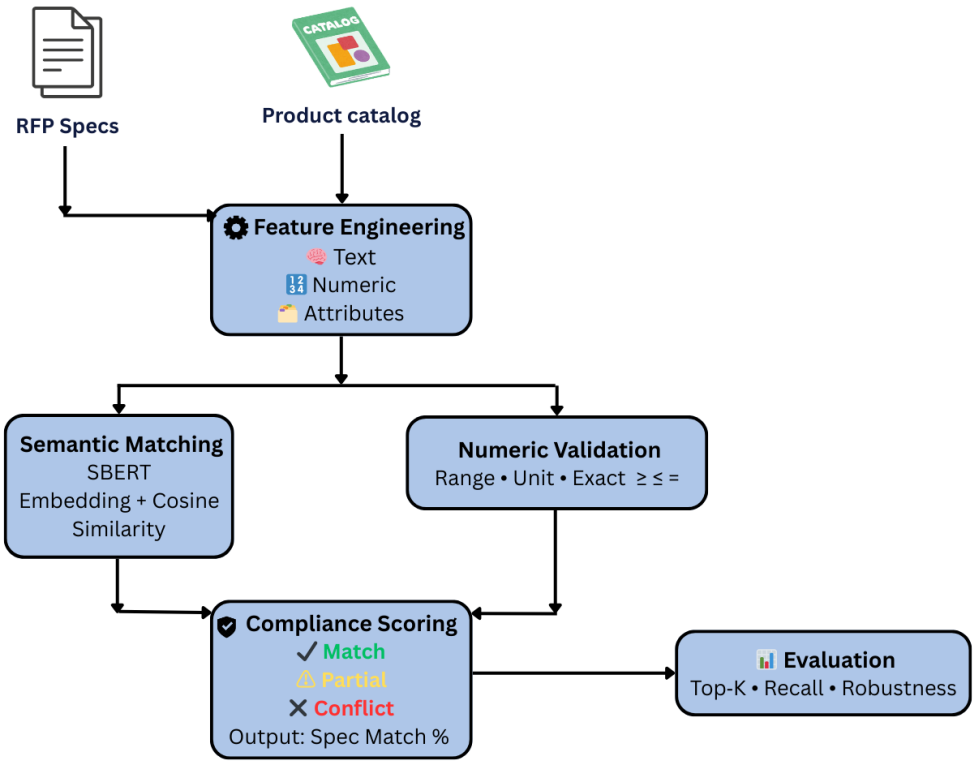

Figure 1: Hybrid RFP–SKU matching pipeline combining semantic similarity (SBERT) and numeric validation for compliance scoring and evaluation.

9 Implementation results

9.1 Embedding Performance

The Sentence-BERT (all-mpnet-base-v2) embedding generation time was:

- **SBERT Embedding Time: 10.31 seconds**

This includes embedding of all RFP specification texts and product catalog entries.

9.2 Compliance Score Distribution

The compliance score distribution across all RFP–SKU pairs (200,000 total comparisons) is summarized below:

Metric	Value
Count	200000
Mean	0.3683
Std Dev	0.1463
Minimum	0.0519
25th Percentile	0.2577
Median	0.3381
75th Percentile	0.4508
Maximum	1.0000

Table 2: Compliance Score Statistics

The distribution shows a right-skewed behavior, with most compliance scores concentrated between 0.2 and 0.5, and a smaller number of highly compliant matches approaching 1.0.

9.3 Structured vs Hybrid Model Comparison

Model	Top-1	Top-K	Recall@K	MRR	Precision@K
Structured-only	1.0000	1.0000	1.0000	1.0000	0.2000
Hybrid (0.5,0.3,0.2)	0.3720	0.5780	0.5780	0.4798	0.1156

Table 3: Comparative Performance of Structured and Hybrid Models

Since the ground truth was generated using structured-only matching, the structured baseline achieves perfect agreement by definition. The hybrid model shows partial deviation, reflecting the influence of semantic similarity and standards weighting.

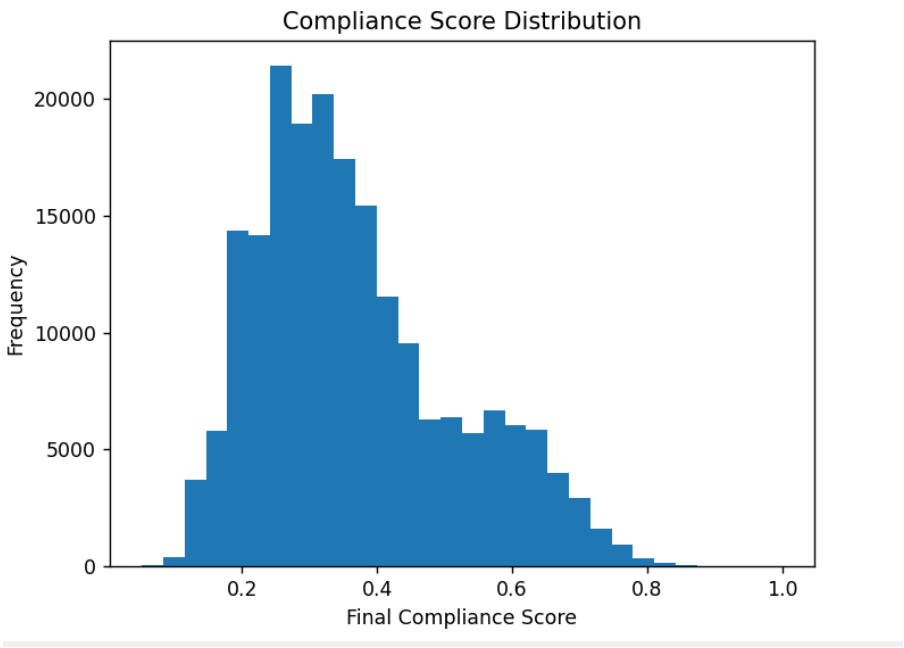

Figure 2: Distribution of Final Compliance Scores

9.4 Ablation Study

To analyze the impact of weight configurations, four combinations were evaluated.

Structured	Semantic	Standards	Top-1	MRR	Top-K
0.7	0.2	0.1	0.4390	0.5977	0.7970
0.6	0.3	0.1	0.4290	0.5839	0.7740
0.5	0.3	0.2	0.3720	0.4798	0.5780
0.4	0.4	0.2	0.3400	0.4384	0.5250

Table 4: Ablation Study Results for Different Weight Configurations

Best configuration:

- Highest Top-1 Accuracy: (0.7, 0.2, 0.1) $\rightarrow$ 0.4390
- Highest MRR: (0.7, 0.2, 0.1) $\rightarrow$ 0.5977

9.5 Weight Sensitivity Analysis

Correlation between semantic weight and performance:

- Semantic weight vs Top-1 Accuracy: $r = -0.859$

- Semantic weight vs MRR: $r = -0.833$

The strong negative correlation indicates that increasing semantic weight reduces agreement with the structured-derived baseline. This suggests that structured constraints dominate technical compliance, while semantic similarity acts as a secondary refinement mechanism.

9.6 Interpretation

The ablation results demonstrate that:

- Structured features should carry the highest weight in engineering specification matching.
- Semantic similarity improves contextual alignment but can override strict structured constraints if over-weighted.
- The optimal balance was achieved at $(0.7, 0.2, 0.1)$.

These findings validate the hybrid architecture design, where deterministic engineering constraints form the backbone of compliance scoring, and transformer-based semantic similarity enhances contextual robustness.

10 Technology Stack

This section describes the software tools, frameworks, and technologies used for the implementation of the proposed spec-level RFP-to-SKU semantic matching system.

10.1 Programming Language

- **Python** – Primary language used for model development, data preprocessing, semantic matching, and evaluation.

10.2 Semantic Matching Framework

- **Sentence-Transformers (SBERT)** – Used to generate dense vector embeddings for RFP clauses and product specifications.
- **HuggingFace Transformers** – Provides transformer-based language models for embedding generation.
- **Tensorflow** – Backend deep learning framework used for implementing transformer models.

10.3 Similarity Retrieval

- **Cosine Similarity** – Used for measuring semantic similarity between RFP clauses and SKU embeddings.
- **FAISS (Facebook AI Similarity Search)** – Used for efficient Top-K vector retrieval.

10.4 Numeric Constraint Validation

- **Custom Rule-Based Engine** – Implements numeric comparison logic including $\geq$, $\leq$, exact match, and range validation.
- **Unit Normalization Module** – Handles unit conversion (e.g., kW $\leftrightarrow$ W).

10.5 Data Processing & Storage

- **Pandas & NumPy** – Used for dataset structuring, preprocessing, and numeric operations.
- **MongoDB** – Used for storing product catalog data, RFP specifications, embeddings, and compliance results.

10.6 Evaluation & Metrics

- **Scikit-learn** – Used for computing evaluation metrics.
- **Evaluation Metrics:**
  - Top-K Accuracy
  - Recall@K
  - Spec-level Compliance Accuracy

10.7 Development Tools

- **Visual Studio Code** – Development environment.
- **Git & GitHub** – Version control and project management.

The selected technology stack ensures modularity, scalability, and efficient implementation of a hybrid semantic and structured compliance matching framework.

11 Project Plan Timeline

Week 1 *Jan 19 - Jan 25*

The research problem was finalized and the spec-level compliance framework was defined. Dataset schemas for `product_catalog` and `rfp_specs` were locked. The overall 7-stage system architecture and system data flow were designed. Repository setup and MongoDB schema configuration were completed.

Week 2 *Jan 26 - Feb 1*

A realistic product catalog dataset consisting of approximately 200 SKUs was generated. Numeric correlations were enforced to ensure engineering consistency. The `product_catalog.csv` file was finalized. A PDF upload endpoint was implemented, documents were stored in the database, and the storage schema was finalized.

Week 3 *Feb 2 - Feb 8*

Approximately 1000 structured RFP clauses were generated. Structured noise was injected to simulate realistic procurement conditions. Engineering realism was validated. PDF text extraction and OCR fallback mechanisms were implemented. Clause segmentation logic was developed.

Week 4 *Feb 9 - Feb 15*

Numeric comparison rules were implemented. The compliance percentage formula was defined. Spec-level labels were generated and train/test splits were created. A clause-to-SKU mapping interface was developed and ground-truth labels were stored in the database.

Week 5 *Feb 16 - Feb 22*

Sentence-BERT / MiniLM was implemented for semantic matching. Clause and SKU embeddings were generated. Cosine similarity and Top-K retrieval mechanisms were implemented. The embedding module was integrated into the Technical Agent and embeddings were stored in the database.

Week 6 *Feb 23 - Feb 27*

Per-spec comparison logic was implemented and compliance thresholds were defined. Compliance percentage computation was completed. Initial accuracy and precision evaluation was performed. The scoring module was integrated into the Technical Agent.

Week 7 *Feb 28 - Mar 8*

Numeric values were extracted from both RFP clauses and SKU specifications. Units were identified

and a numeric comparison model was designed. The numeric parser was integrated into the Technical Agent to enable structured constraint handling.

Week 8 *Mar 9 - Mar 15*

Unit conversion logic (e.g., kW $\leftrightarrow$ W) was implemented. $\geq$, $\leq$, and between-range comparison logic was developed. Tolerance handling was added and numeric logic was fully integrated into the compliance scoring pipeline.

Week 9 *Mar 16 - Mar 22*

Semantic similarity scores were combined with numeric compliance scores. The compliance percentage formula was refined. Numeric correctness rate was evaluated and compliance gaps were stored in the database.

Week 10 *Mar 23 - Mar 29*

Partial compliance detection was implemented. Spec-wise error patterns were analyzed. Top-3 alternative SKU suggestions were added and compliance gap reporting was implemented.

Week 11 *Mar 30 - Apr 5*

The system was evaluated under missing parameters, ambiguous clauses, paraphrased clauses, and unit inconsistencies. Performance degradation under different noise types was measured and fallback logic was improved.

Week 12 *Apr 6 - Apr 12*

Detailed error analysis was conducted. Threshold tuning and attribute weighting improvements were performed. Embedding caching was optimized and retrieval performance was enhanced.

Week 13 *Apr 13 - Apr 19*

Final evaluation metrics were computed, including spec-level accuracy, numeric correctness, compliance percentage error, and robustness delta. An ablation study was conducted and the final reporting module was generated.

Week 14 *Apr 20 - May 8*

The methodology, dataset, and results sections were documented. Tables and figures were prepared. The demo version was finalized and complete system testing was performed prior to submission.

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12 Conclusion

This research addressed the problem of clause-level semantic matching between industrial RFP technical requirements and OEM product catalog specifications. Traditional procurement processes rely heavily on manual evaluation of technical clauses, making the system inefficient, inconsistent, and prone to human error. Existing semantic similarity approaches focus primarily on document-level comparison and do not provide structured, spec-wise compliance validation required in engineering procurement workflows.

In this phase of the project, the primary objective—designing and implementing a spec-level semantic matching framework—has been successfully completed. A structured RFP–SKU dataset was constructed by aligning engineering attributes extracted from RFP clauses and corresponding product catalog specifications. The dataset incorporates semantic, numeric, and categorical attributes such as voltage rating, core configuration, cross-sectional area, insulation type, fire resistance, and standards compliance.

A transformer-based sentence embedding model (Sentence-BERT / MiniLM) was implemented to generate dense vector representations of RFP clauses and product specifications. Using cosine similarity and Top-K retrieval mechanisms, the system identifies the most relevant candidate SKUs for a given requirement. The implemented pipeline includes preprocessing, feature structuring, embedding generation, similarity computation, and a preliminary compliance scoring module that produces attribute-wise match/non-match results along with an overall compliance percentage.

The experimental results demonstrate that semantic embeddings effectively capture technical requirement similarity at the clause level. The Top-K retrieval performance indicates that relevant SKUs are consistently ranked among the highest candidates, validating the effectiveness of embedding-based semantic matching for structured engineering data. The implemented compliance scoring mechanism further enables interpretable outputs, allowing users to identify matched attributes and potential mismatches rather than relying solely on raw similarity scores.

The system has been developed as a modular, extensible framework, ensuring scalability for future integration of numeric constraint reasoning and robustness analysis. The current implementation establishes a strong foundation for transitioning from document-level similarity systems to structured, spec-level compliance-aware procurement automation.

Overall, the completed phase confirms the feasibility and effectiveness of a semantic embedding-based approach for spec-level RFP-to-product matching. The framework provides a solid basis for further enhancements, including numeric constraint validation, robustness evaluation under incomplete specifications, and advanced explainability mechanisms in subsequent project phases.

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